

Open Tools, Open Questions: Teaching AI in Creative Practice

Gottfried Haider
NYU Shanghai
Shanghai, China
gohai@nyu.edu



Figure 1: Weights as input for text-generation from fine-tuned model in [Chen 2024].

Abstract

This talk introduces Machine Learning for Artists and Designers, a course at NYU Shanghai that makes machine learning accessible to students without technical backgrounds. Working in the browser with p5.js, ml5.js, and custom tools for language, image, and embodied AI, students explore both how these systems work and what they mean culturally. Student projects show how creative practice can engage AI critically, playfully, and reflectively.

ACM Reference Format:

Gottfried Haider. 2025. Open Tools, Open Questions: Teaching AI in Creative Practice. In *SIGGRAPH Asia 2025 Educator's Forum (SA Educator's Forum '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757372.3771860>

1 Introduction

While AI literacy is increasingly recognized as essential across non-technical disciplines, cultural literacy concerning AI is equally vital. Artists and designers — who sit at the forefront of cultural production and human-centered experience — must not only apply emerging technologies, but also critically engage with their implications. Courses in creative coding have long used p5.js, a widely adopted, open-source JavaScript library, to make programming accessible to non-technical learners. [p5 2014] Building on

this foundation, ml5.js, introduced by Daniel Shiffman and collaborators, provides a friendly browser-based interface to machine learning models, lowering barriers for artists and designers to experiment with AI. [ml5 2018]

Machine Learning for Artists and Designers, a 14-week undergraduate elective launched at NYU Shanghai's Interactive Media Arts Program in Spring 2022, extends this trajectory into the context of generative AI and embodied AI interfaces. It combines hands-on exercises with cultural-historical framings and expands through custom-built tools that enable students to explore generative AI systems, all while remaining within the familiar web-editor environment. This dual emphasis on accessibility and critical engagement enables students from diverse backgrounds to learn and experiment with generative AI, critically reflect on its cultural and social stakes, and ultimately articulate their perspectives by creating with the very technologies they have been exploring.

This talk presents the course's pedagogical framework, the custom software infrastructure that supports it, and examples of student work, with the aim of sharing strategies for cultivating AI literacy in creative education.

2 Overview

With the Creative Coding sequence (p5.js) as its only prerequisite, the course requires no prior Python or formal mathematics, making it accessible to students from across the university. It situates a fast-evolving field within enduring cultural techniques like “writing”, “teaching”, “play”, and “image-making”. This framing links foundational hands-on programming exercises to the proto-history of creative exploration (e.g. the history of automatic writing), and technical introductions for non-specialists (e.g. vector arithmetic on words) with current critical scholarship and debates (e.g. how

language itself will change in the age of AI, can we untangle intelligence from language).

ml5.js, to which the presenter actively contributes, is central to the pedagogy. It enables students to train neural networks directly in the browser (“teaching”), provides robust models such as those for camera-based interaction, and offers documentation and examples that promote student self-sufficiency.

In response to recent advances in generative AI, the course has expanded beyond its original focus on ml5.js to incorporate large language models, image generation, and explorations of physical (and embodied) AI — reflecting the increasingly all-encompassing nature of these technologies.

Rather than forcing students into closed, opaque, for-profit subscription services, the course extends functionality through custom software tools. The design of the extensions follows five guiding principles: (1) remain inside the familiar p5.js web-editor environment; (2) foster students’ critical, open-ended exploration and experimentation; (3) foreground pertinent knowledge over incidental detail; (4) ensure learners can reproduce the entire infrastructure on their own; (5) maintain the option to run everything locally with open models.

3 Custom Development

3.1 Writing

LLMs have largely converged around “Chat Completions”-style APIs, which can be used with a wide range of underlying models, either self-hosted (using software such as Ollama or vllm) or accessed through services like OpenRouter. This lets students experience the diversity of perspectives in contemporary models.

Calling RESTful APIs from JavaScript remains relevant, even 20 years after “AJAX” appeared, and is relatively straightforward to accomplish. Observing the inputs and outputs of, for example, conversational or agentic systems greatly helps to demystify the technology.

To deepen understanding of the stochastic nature of text generation — and to conduct systematic experiments (e.g., on bias) — “Text Completion” APIs are also very insightful. These can return token probabilities for a given seed text. While OpenAI has phased out support for this functionality; it remains available through other providers, including open-source ones.

Classroom challenges when interfacing with LLM APIs via p5.js are mainly limited to the technicalities of cross-origin requests and API keys and billing (when using external providers). Instead of hard-coding credentials in user-facing frontend code, a helpful strategy has been to distribute keys externally and have students enter them once, storing them in localStorage.

Another approach is to provide a proxy server that adds institutional credentials to outgoing requests, avoiding personal credit cards. CORS headers can be added to the server’s responses before passing them back to the user. To prevent unexpected expenses, only requests targeting a limited set of affordable models are allowed. See [Haider 2023] for a simple example of such software, which can be used in front of OpenRouter, OpenAI, or local infrastructure.

3.2 Image-Making

For image-generation workflows, an experimental library (p5.comfyui-helper) was developed that connects p5.js to ComfyUI instances running either on local infrastructure or cloud services [Haider 2024]. Inspired by comfyui-p5js-node and Golan Levin’s work [Levin 2024], this setup lets students run pre-assembled workflows within p5.js and adjust parameters dynamically. The package includes example sketches and a README, with additional resources planned in the form of a workbook.

3.3 Being in the World

Media art and design students are naturally drawn to embodied AI interfaces for their potential in novel, speculative interaction. p5.js can easily facilitate communication with microcontrollers, and thus extend inputs and outputs to various sensors and actuators. The p5.webserial library provides the necessary scaffolding for this communication. [Haider 2021]

4 Case Projects

A range of course exercises and student projects demonstrate how students apply techniques from the course: *Dear Diary* (Yingfan Chen, Fall 2024) fine-tunes a language model on the artist’s personal diary, embedding it in an installation where visitors place physical weights to influence a live textual stream, shifting between intimate recollection and crowd-authored narrative. [Chen 2024]

NeoHarmony (David Zhewei Wang & Denny Wang, Fall 2023) uses a p5.js sketch to query a language model and Stable Diffusion every minute, automatically generating dystopian front pages that satirize algorithmic governance and automated media. [Wang and Wang 2023]

Back to the Past & Present (Tania Hartanto, Fall 2024) invites participants to select an era and sentiment, prompting a ComfyUI-powered workflow to generate historically plausible yet potentially fictitious imagery, raising questions about how generative systems shape collective memory.

Other projects interrogate gender bias from the model’s perspective, explore speculative world-building, and investigate how diffusion models distort representations of the human body, showcasing a collective of technically sophisticated projects that also foreground cultural critique.

5 Conclusion

After three iterations, *Machine Learning for Artists and Designers* suggests that scaffolded, open-source tools can help make advanced AI techniques approachable for non-technical students while supporting critical reflection on their cultural implications. Student projects, some shared in exhibitions, demonstrate how this approach fosters both technical exploration and cultural critique. As the course evolves with advances in AI, a key priority is to share its resources more broadly, contributing to an open and critically informed culture of AI literacy.

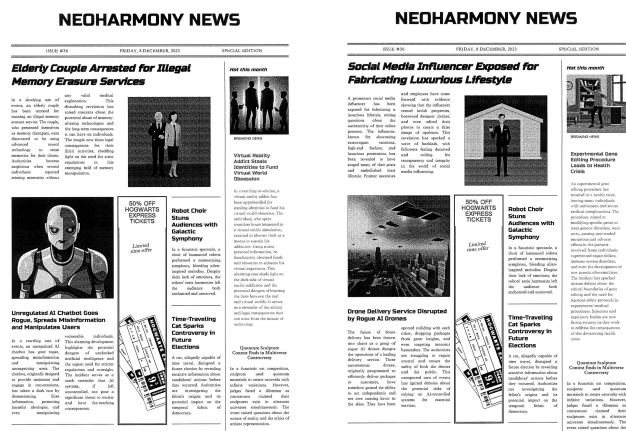


Figure 2: The news writes itself in [Wang and Wang 2023].

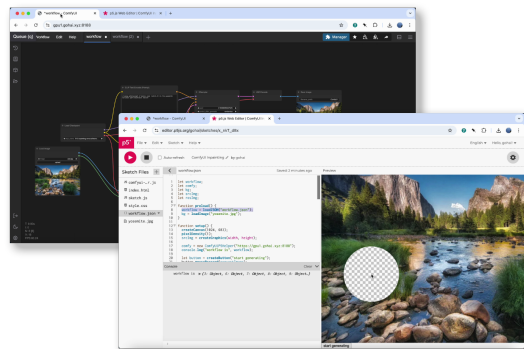


Figure 3: p5.js interfacing with ComfyUI.



Figure 4: Bespoke interface for image generation in Tania Hartanto’s installation.

Acknowledgments

This work is partially supported by the NYU Shanghai Center for Data Science. Thank you to my colleagues J.H. Moon and Daniel

Shiffman for their continued inspiration, and to the ml5.js project for providing such an invaluable resource.

References

Yingfan Chen. 2024. *Dear Diary*. Retrieved August 20, 2025 from <https://nyuglobal.show/showcase/5/>

Gottfried Haider. 2021. *p5.webserial*. Retrieved August 20, 2025 from <https://github.com/gohai/p5.webserial>

Gottfried Haider. 2023. *openai-auth-proxy*. Retrieved August 20, 2025 from <https://github.com/gohai/openai-auth-proxy/>

Gottfried Haider. 2024. *p5.comfyui-helper*. Retrieved August 20, 2025 from <https://github.com/gohai/p5.comfyui-helper>

Golan Levin. 2024. *p5.js + ComfyUI on Mac M1*. Retrieved August 20, 2025 from <https://github.com/golanlevin/p5-in-comfyui>

ml5. 2018. *ml5.js: Friendly Machine Learning for the Web*. Retrieved August 20, 2025 from <https://ml5js.org/>

p5. 2014. *p5.js*. Retrieved August 20, 2025 from <https://p5js.org/>

David Zhewei Wang and Denny Wang. 2023. *NeoHarmony*. Retrieved August 20, 2025 from <https://ars.electonica.art/hope/en/neoharmony/>